

Smart Meal Planner

Optimized User Flow & Architecture (Hybrid Model)

Core Navigation (4 Tabs)

- 1. Discover — Inspiration layer (short-form recipe feed).
- 2. Recipes — Structured recipe database (manual + imported).
- 3. Plan — Weekly calendar & meal scheduling.
- 4. Profile — Preferences, dietary settings, account.

Primary User Paths

- Mode A (Inspiration First): Discover → Save → Recipes → Plan → Grocery List.
- Mode B (Utility First): Recipes → Plan → Grocery List.

Discover Flow

- Scroll vertical recipe feed.
- Search & browse categories.
- Open recipe preview.
- Actions: Save / Add to Plan / View Details.

Save & Structuring Gate

- User confirms structured recipe data.
- Editable ingredients (quantity + unit).
- Editable steps.
- Auto-suggested tags (user editable).
- Nutrition engine runs after save.

Recipes Tab

- Search & filter saved recipes.
- Collections & tagging.
- Manual recipe creation.
- Future: AI import from URL.
- Recipe detail includes nutrition breakdown.

Plan Tab

- Weekly calendar grid view.
- Add recipe to day.
- Adjust servings per instance.
- Display daily & weekly nutrition totals.
- Generate grocery list.

Grocery List (Derived View)

- Aggregated ingredients from weekly plan.
- Serving multipliers applied.
- Grouped by category (Produce, Pantry, etc.).
- Check-off items.
- Export/share functionality.
- Future: Pantry-aware filtering.

System Architecture Alignment

- Discover does not own structured data.
- Recipes are canonical structured entities.
- MealPlanEntries reference Recipes.
- Grocery list is derived projection, not independent entity.